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In this age of cyber crime and fraud, are you sure of your safety? Are you safer online or in a shopping mall?

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How do you stay safe despite your exposure to harmful viruses out there?

INSECURE AIRWAVES

After the series of terror threats and attacks emerging out of unsecured wireless networks, it's high time we learnt our lessons. We take a look at the state of our society when it comes to secure technology

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Wireless fidelity (Wi-Fi) penetration is growing by leaps and bounds with, not only business houses, but also a lot of residential users opting for this facility. As most households nowadays, have a laptop along with a desktop, it makes sense to have a Wi-Fi router within homes for flexible internet access. As with the internet, proliferation breeds vices. If the number of cyber crimes occurring in the country is any indication to go by, the possibility of unsecure Wi-Fi networks being compromised is also high.

Just a year ago, an American citizen, Kenneth Haywood, residing in Navi Mumbai didn't secure his Wi-Fi connection. The result — hackers accessed Mr Haywood's account to send a terror email. What followed made front page news in the city's dailies. Police personnel carried out mock Wi-Fi security checks. Awareness campaigns were launched to inform people to secure their Wi-Fi networks. But till today

that message has somehow not really penetrated. The next victim is just around the corner. Unsecure Wi-Fi networks can make you lose much more than just your internet bandwidth and money.

Digit Wi-Fi security test

We went to societies in and around our office to check for open unsecure networks. We used a Gateway netbook for the test which has an internal Wi-Fi network adapter. We did not hack into any account by second guessing the user name / passwords or used any tools to gain entry into a Wi-Fi hotspot. The test was very basic where we just used our netbook to check for any open unsecure network. Once an unlocked network was recognised, we opened it. After surfing the web for some time to check if the security of the network could be breached, we disconnected.

As expected we came across two open wireless networks having speeds upto 54Mbps. These networks had absolutely no security. They showed up as unsecured networks when the wireless connections page was refreshed. So



many people are still ignorant of security regarding their wireless networks. A study conducted by the Data Security Council of India and Deloitte states that over 80 per cent of the wireless networks across the country (out of the 35,000 networks tested) are vulnerable to attacks.

Encryption woes

Data transfer over wireless networks is in the air and not confined to any wire. Due to this it is easy for a hacker to snoop into the information that is being transferred over the wireless network. To prevent this there is a need to encrypt the data in a way that it can only be deciphered by someone who has the decryption key.

Now the wireless routers have two kinds of encryption: Wired Equivalent Privacy (WEP) and Wi-Fi Protected Access (WPA). WEP is considered to be a weak encryption mechanism as it has many loopholes. It uses 64-bit or 128-bit security keys to encrypt data of which 24-bits are called Initialisation Vector (IV hereon). Since 24-bits give only 16.7 million variations, it is comparatively easier to decode the keys. Also the possibility of using the same IV key more than once is high making the system vulnerable to hack attacks.

WPA on the other hand is considered much safer than WEP because it has a 48-bit IV which gives over 500 trillion variations, which is very difficult to crack. Also the IV keys are better



A hotspot close to our office where we found an open, unprotected Wi-Fi network